

Predicting NHL Goaltender Performance: A Machine Learning Approach to Starting Goaltender Decisions

Katie Wimmer, Sam Sieloff, Nick Borchich & Kevin Taffou

School of Data Science and Society – MADS

University of North Carolina

Chapel Hill, NC

`krwimmer@unc.edu, ssieloff@unc.edu`

`borchich@unc.edu, ktaffou@unc.edu`

1. Introduction

In professional hockey, deciding which goaltender to start on a given night is one of the most important choices a coaching staff makes. The difficulty of these decisions was recently highlighted during the 2026 Milan Cortina Winter Olympics, where the goaltending decisions for the top contenders dominated the headlines before any games were even played. These all-star teams had several rostered goalies who were already the best in the world, and it was a major debate as to which out of an elite group would get to start in the important games. While this Olympic situation is uniquely tense, decisions like this are made every night in the National Hockey League (NHL), where each team carries three to four goalies throughout the season.

Traditional metrics like Goals Against Average (GAA) and save percentage give a basic insight into goaltender performance, but they do not account for the difficulty of the matchup, how rested a goaltender is, or other game factors like home-ice advantage or playoff pressure. A goaltender with a strong GAA may have simply faced weaker opponents, while another with a worse GAA may have consistently been put up against the league's best offenses. These limitations make traditional statistics an incomplete basis for starting goaltender decisions.

This project builds a machine learning model to predict how many goals a goaltender will allow in a given NHL game, using only data available before the game is played. Instead of using raw

season averages, the model creates feature vectors for both the goaltender and the opposing team that weigh recent performance more heavily than older games, exploring different averaging and smoothing methods to accomplish this. These features are combined with game-context information such as home or away status, recent win and loss streaks, and whether the game is in the regular season or the playoffs.

2. Related Work

Prior work applying machine learning to NHL prediction has primarily focused on game-outcome prediction rather than individual goaltender performance. Major projects have tried to predict whether a team will win or lose. Weissbock et al. in 2013 were among the first to apply machine learning to forecast NHL game winners, finding that neural networks achieved approximately 59% accuracy. That result has since appeared to represent an upper bound for win prediction. Because hockey is an incredibly random game and susceptible to upsets, an accuracy of 59% was considered impressive.¹

More recent work has explored rolling and momentum-based features for predicting the winner of a game. These additions accounted for more factors than Weissbock et al. and made compelling arguments for incorporating momentum and in-season form. We drew some inspiration from these approaches and focused on rolling averages and recency-weighted statistics over traditional frequency-based statistics like save and shot percentages. In terms of goalie-specific statistics that could be utilized, there have been advanced metrics such as Goals Saved Above Average (GSAA) and Goals Saved Above Expected (GSAX).² These metrics adjust for shot quality and team context, but they are calculated retroactively after a game is played. These advanced goalie statistics are performance evaluations rather than predictive models, and they do not incorporate the specific matchup against an opposing team, recency weighting, or workload information. They proved to be too conceptually distinct from the other metrics we ultimately used, though they hold potential for future related projects.

Our work builds upon prior NHL predictive tasks by constructing a goaltender-specific predictive framework that uses both offensive and goaltending statistics. It focuses on the matchup between a goalie and an opposing offense rather than simply predicting game outcomes from aggregate historical data. An equally important distinction is that none of our computed statistics were calculated retroactively, which is discussed in greater detail in the Methods section.

¹Weissbock, Josh & Viktor, Herna & Inkpen, Diana. (2013). *Use of performance metrics to forecast success in the NHL*.

²Naples, Marc; Gage, Logan; and Nussbaum, Amy (2018). "Goalie Analytics: Statistical Evaluation of Context-Specific Goalie Performance Measures in the National Hockey League," *SMU Data Science Review*: Vol. 1, No. 2, Article 12. Available at: <https://scholar.smu.edu/datasciencereview/vol1/iss2/12>.

3. Methods

3.1 Data Collection

Play-by-play data was collected via the NHL Stats API (`api-web.nhle.com/v1`) for five seasons from 2019 through 2023, covering both regular season and playoff games. We chose to limit the window to five years due to concerns about processing excessively large datasets and to avoid incorporating statistics from a meaningfully different era of NHL play. The average number of goals scored per game tends to shift over periods longer than five to eight years. With five seasons, the dataset covered over 6,500 games and more than five million events. Events included everything from faceoffs to penalties, but we ultimately filtered to include only shots, saves, and goals. Each event record includes game context (game ID, date, home and away teams, current score), player identifiers for the goalie and shooter, shot type, on-ice strength (5-on-5, power play, or shorthanded), and x/y coordinate location on the rink. After filtering for offensive and goalie-related data, we retained over 1.5 million events.

3.2 Feature Engineering

For each game, we computed a set of statistics using only events that occurred prior to that specific game, ensuring no risk of data leakage. For each goaltender appearance we calculated save percentage overall as well as broken down by game situation (even strength and power play), by period, across twelve five-minute windows throughout regulation, by each of the eleven shot types in the data (wrist, snap, slap, tip-in, backhand, deflected, wrap-around, poke, bat, between-legs, and cradle), and for high-danger shots taken from the slot area directly in front of the net (defined as $|x| > 70$, $|y| < 20$ in standard NHL rink coordinates). If a goalie faced no shots in a particular category in a given game, that value was left as missing rather than recorded as zero or one, which would be misleading. Altogether we considered 158 different statistics, which were later reduced to a much smaller set. The complete list of metrics considered is provided in the appendix.

3.3 Recency Weighting

Rather than treating all past games equally, we applied two different methods for weighting recent games more heavily when aggregating statistics into a pre-game summary vector for each goaltender.

The first method was exponential decay weighting, in which recent games contribute more heavily to the summary. The weight assigned to a game played Δ games ago is:

$$w(\Delta) = 2^{-\Delta/30} \tag{1}$$

This means a game played 30 games ago carries half the weight of the most recent game, a game played 60 games ago carries one-quarter the weight, and so on. This scheme produced our exponentially weighted mean (EWM) statistics.

In addition to the EWM, we computed a simple 10-game rolling mean as a secondary recency-weighting approach — the arithmetic average of each statistic over the previous 10 games. Rolling features were included only once a goaltender had played at least four prior games. The rolling mean therefore captures short-term form more sharply, while the EWM provides a smoother long-run estimate. Including both gives the model the flexibility to leverage recency at different time scales.

3.4 Additional Features

Beyond save-percentage-based features, the feature vectors included workload indicators such as days of rest since the last game and the number of games played in the last 7, 14, and 30 days. We also added stability measures, including the rolling 10-game standard deviation of save percentage and goals allowed. A high standard deviation indicates a volatile goaltender whose performance is difficult to predict.

The final goaltender feature vector contains 77 columns. The opponent offensive feature vector, constructed using the same EWM and rolling-average approach applied to the opposing team’s shooting statistics, contains 78 columns. Combined, the training dataset has 13,284 rows covering 168 goalies across 6,232 games. The total of 158 features before selection includes game-context columns such as home or away status.

3.5 Feature Selection via Elastic Net

To reduce the initial 158-feature space to a manageable and interpretable set, we applied an Elastic Net regularization model, which combines L1 and L2 penalties to simultaneously shrink and eliminate less informative features. This process selected 31 features that aimed to identify the most predictive features while discarding redundant or noisy statistics. The selected features include both goaltender and opponent characteristics, and it chose a mix of exponentially weighted means (EWM) and 10-game rolling averages. The fact that both recency-weighting schemes survived selection together confirms that they both contained useful information. The rolling mean captures more recent data while the EWM encodes a smoother longer-run trend. Between the two of them, the model was able to benefit from having access to both time scales rather than just one.

Several patterns in the selected features are worth highlighting. First, save percentage emerged as the single most important class of feature, appearing across multiple game situations, time windows, and shot types. More interestingly, some features that were highlighted are not uniform across the game. For example, third-period save percentage and several specific five-minute windows (such as

minutes 10–15, 40–45, and 55–60) were selected, suggesting that performance in particular phases of a game is more predictive than overall averages. Second, certain shot types were selectively important depending on if they were regarding goaltenders or offenses. For example, tip-in save percentage was retained as a goaltender feature, while backhand and deflected shot percentage appeared on the opponent side. A more hockey technical take on this can suggest that tip-ins are a challenging to save and a known weakness for some goaltenders. Or that a team that generates a high backhand shooting percentage may have a more overall dangerous offensive attack. Finally, opponent statistics such as even-strength and power-play shots for, high-danger shots for, and career shooting percentage round out the selected set, confirming that the quality and quantity of the opposing offense is just as important as the goaltender’s recent stats. The list of the 31 metrics selected are also in the appendix.

4. Experiments and Results

The dataset was split chronologically to prevent look-ahead bias, with 80% of games (ordered by date) used for training and the most recent 20% used for testing.

4.1 Goal Prediction Results

The Linear Regression and Random Forest models both struggled to predict the exact number of goals allowed. The Linear Regression model achieved a Mean Squared Error (MSE) of 2.47 on the test set. When predictions were rounded to the nearest integer, the model correctly predicted the exact number of goals only 22.2% of the time. However, when evaluating within a tolerance of ± 1 goal, accuracy improved substantially to 63.5%. The Random Forest model achieved an MSE of 2.56 and an R^2 value of -0.04 , indicating that a simple mean of the training labels would be a better predictor. Its accuracy within one goal was 63.4%.

These results suggest that while the models can capture a general range of performance, they cannot reliably predict the exact outcome due to the high variance of hockey scoring. The majority of predictions fell between three and four goals, which aligns with the distribution of the target variable, whose mean is 2.75 goals.

4.2 Binned Classification Results

The multinomial logistic regression achieved a multiclass accuracy of 23.0% across six bins. The confusion matrix revealed that the model predominantly predicts the central bins of two, three, and four goals. While not highly accurate, the model demonstrates skill in capturing the central tendency of the data.

Under a relaxed accuracy metric requiring predictions to fall within ± 1 bin of the true value, performance improved to 63.6%, closely aligning with the within-one-goal accuracy observed in the regression models. Misclassifications were most common between adjacent bins (2 vs. 3 goals, 3 vs. 4 goals), while errors spanning two or more bins were rare. This pattern suggests the model reliably identifies the general tier of goaltender performance even when it cannot pinpoint the exact goal total.

4.3 Pairwise Comparison Results

The pairwise comparison model, which directly compares two goaltenders scheduled to start against the same opponent, achieved an accuracy of 58.9% on the test set. This represents a significant improvement over the baseline accuracy of 50.0% for random guessing and the “always pick goalie 1” baseline of 50.8%. The precision for predicting that goalie 1 would perform better was 0.59, the recall was 0.61, and the F1-score was 0.60. The most important features for this model were the exponentially weighted save percentage metrics, indicating that recent form is the strongest predictor of future comparative performance. The top ten most important features included exponentially weighted save percentage, exponentially weighted even-strength save percentage, career save percentage, rolling high-danger save percentage, and various period-specific save percentages.

The model’s ability to achieve nearly 60% accuracy on a binary task that is inherently noisy demonstrates its practical utility. While perfect validation is impossible since two different goalies can never face the exact same opponent under the exact same conditions, the pairwise model provides a robust method for comparative goaltender selection.

5. Conclusion

5.1 What We Learned

While the project overall has the potential to provide value to coaching staffs and analysts, noise continues to be the dominant factor in goaltender performance prediction. This is ultimately not surprising given the nature of professional sports, particularly one with an 84-game regular season. If noise were not the dominant factor, the game would be far less exciting and compelling to watch. The plurality of NHL games are decided by a single goal, so even if noise dominates, a slight improvement in goals allowed can easily be the difference between winning and losing. Playoff races are often extremely tight, and a strategy that produces even a few additional wins over the course of a season could still be practically valuable.

It is also worth reemphasizing what the core goal of this project was: not to predict the exact 6

score of a game, but to determine which of two goaltenders, placed in the same situation against the same opponent, is more likely to allow fewer goals. Looked at this way, the pairwise comparison model's accuracy of approximately 60% is actually a meaningful result. In any binary decision where the two options are genuinely difficult to distinguish and the underlying outcome is heavily influenced by randomness, consistently being right 60% of the time could be a big advantage. It could also alleviate concerns over bias in goaltender decisions. Over a full 82-game season, that difference compounds. Coaching staffs are not looking for exact certainty in prediction for number of goals they are looking for any systematic advantage they can find, and a model that is right more often than not on a genuinely hard problem is a meaningful step in that direction

5.2 Potential Contributions to the Field

Sports analytics is a constantly evolving field. Despite professional baseball being well over 100 years old, the use of analytics continues to change with ongoing developments in sabermetrics. Until relatively recently, walks were perceived as far less valuable than base hits; we now understand that reaching base is what matters most, regardless of how it is accomplished. The importance of relief pitchers as strategic assets rather than last resorts is a concept that is recent relative to the history of the sport. Even today, the measurement of closer effectiveness and optimal deployment remains an active area of debate. Similarly, the perceived value of running backs in American football has shifted repeatedly in recent years, and the football community has broadly come to accept that the quarterback's contribution is so dominant that it is nearly impossible for another position player to win the league's Most Valuable Player award.

The point is that while the current consensus in hockey is to maintain a goaltender rotation and start the best available goalie whenever possible, it is entirely plausible that this view will evolve. NHL teams operate within a salary cap, so any data-driven edge they can gain is welcomed, even if the marginal impact on win percentage appears modest. This includes the potential to be more selective and intentional about goaltender starts and rotation management.

6. Future Work

There are several directions in which this project could be extended. The current models do not account for gaps between seasons; the final games of one season blend directly into the early games of the next, which may introduce noise into recency-weighted features. A principled method for resetting or discounting cross-season statistics would be a valuable improvement.

Expanding the dataset to seasons prior to 2019 could improve the stability of features that aggregate further into the past, though there are real risks associated with incorporating data from a

meaningfully different era of play. Using event data from the early 2000s to predict games in 2026 would likely introduce distributional mismatch, but the tradeoffs would be worth exploring. The scale of event data from such an expansion would also pose computational challenges.

Another avenue for improvement would be incorporating the advanced metrics discussed in the Related Work section. Goals Saved Above Expected (GSAx) as either a feature or a prediction target would provide a metric adjusted for shot quality rather than just shot type, potentially offering a cleaner signal of true goaltender skill. This metric has also been receiving considerably more attention in hockey media and front offices in recent years.

Shot and skater speed are also not currently modeled. The NHL has deployed new tracking systems capable of recording puck and player speed in real time, and incorporating this information could enrich the feature set substantially. Finally, one could consider weighting goals by their game-state context. A save made when the score is tied in the third period is far more consequential than one made with a large lead, and the NHL's tracking of game-winning goals could serve as a starting point for quantifying this contextual value.

Appendix

The appendix includes two components:

1. **Complete list of features considered** — all 158 statistics evaluated prior to feature selection, organized by category (save percentage, workload, stability, game context).
2. **List of features selected by the elastic net**
3. **Notebook file with all project code** — the full Python notebook covering data collection via the NHL Stats API, feature engineering, model training, and evaluation.